

Sabin Baral

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EDUCATION

The University of Southern Mississippi

Hattiesburg, MS

Bachelor of Science in Polymer Science and Engineering | Minor: Chemistry

May, 2028

Relevant Coursework: Polymer Rheology, Polymer Mechanics, Problem Solving, Calculus III, Organic Chemistry II

GPA: 4.0/4.0 (President's List)

RESEARCH AND ENGINEERING EXPERIENCE

Undergraduate Researcher and Instrumentation Engineer

Dec 2024 – Present

Gu Research Group

Hattiesburg, MS

- **Autonomous Lab Orchestration:** Architecting a "Self-Driving" materials lab by developing a **Python-based Hardware Abstraction Layer (HAL)** to synchronize an N9 North Robotics arm with Thorlabs motorized stages and Osilla spectrometers, creating a fully **closed-loop experimental cycle**.
- **High-Throughput Polymer Research:** Managing the fabrication and characterization of **1000+** block copolymer-homopolymer thin-film samples; utilizing **robotic spin-coating** and **automated AFM** to map phase behavior and **Order-Disorder Transitions (ODT)** with **40% greater time efficiency** than manual workflows.
- **Hardware Integration & Driver Development:** Authored custom Python drivers to enable **software-defined control** of standalone instruments, ensuring uniform pressure during **electrical probing** and precise **optical metrology** for **film thickness verification**.
- **Advanced Structural Analysis:** Conducting grazing-incidence X-ray scattering (**GISAXS/GIWAXS**) at **in-house facilities** and assisting at **national synchrotron facilities (ALS at LBNL and NSLS-II at Brookhaven)** to interpret **nanoscale structural ordering** in high-throughput sample libraries.
- **Infrastructure Engineering & Design:** Leveraging **OpenSCAD and AutoCAD** to bridge hardware gaps through **additive manufacturing**; engineered **custom robotic interfaces**, high-capacity sample holders (achieving a **500% increase in throughput**), and protective UTM housings that ensure **equipment longevity** and workspace efficiency.

Visiting Research Scholar

Sept 2025

Lawrence Berkeley National Lab

Berkeley, CA

- **Automated Screening:** Utilized automated Atomic Force Microscopy (**AFM**) to characterize surface morphology and phase behavior across **500+ high-throughput thin-film samples**.
- **Database Design:** Collaborated with staff scientists to build a comprehensive sample database, creating a massive data repository to facilitate pre-screening by polymer type.
- **Robotic Integration:** Assembled and optimized a multifunctional robotic system for spin coating and thermal annealing; trained collaborators on operation protocols.

Undergraduate Research Assistant

Oct 2024 – May 2025

Center for Optoelectronics and Devices

Hattiesburg, MS

- **Device Fabrication:** Fabricated **organic solar cells** (p-type conjugated polymers/n-type acceptors) in a nitrogen glovebox, achieving **power conversion efficiencies** in the **15–18% range**.
- **Stability Analysis:** Investigated degradation pathways under heat and light exposure using accelerated heating systems to correlate processing parameters with device stability.
- **Data Interpretation:** Characterized active layers using **UV-Vis, AFM, and I-V** measurements; interpreted complex datasets using Python to optimize donor-acceptor combinations.

PUBLICATIONS

Upreti, S.; Xu, L.; Moniruzzaman, M.; Wang, Y.; Adhikari, K.; **Baral, S.**; Patton, D.; Ma, B.; Xu, J.; Li, R.; Zhu, C.; Xia, W.; Gu, X. "**A Robotic High-Throughput Grid-Search Platform for Mapping Phase Behavior in Triblock Copolymer-Homopolymer Blends.**" *JACS Au* (Under Review).

SKILLS

Characterization: GISAXS, GIWAXS, AFM, DLS, DSC, TGA, FTIR, UV-Vis Spectroscopy.

Programming & Software: Python, MATLAB, Igor Pro, OriginPro, ChemDraw.

Design & Prototyping: AutoCAD, OpenSCAD, Cura

HONORS AND AWARDS

Winner, RoboSoccer Tournament – Google Developer Student Clubs, USM

2025

Valedictorian and Student Body President Scholarship, USM

2024